UC San Diego
RADY SCHOOL OF MANAGEMENT

MGT 450: Project Management

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Note: This syllabus may be updated and revised at a later date.

COURSE DESCRIPTION

As projects become more complex and acquire (never before seen) global characteristics, the demands placed on the project manager and her team are also changing. Whether working for a startup designing and building the next-generation wireless technology or leading a small volunteer effort to organize an open source project, competent project leadership requires understanding how to allocate financial, material, time-based resources, and most importantly how to motivate and maintain focus of your project team. Good intentions cannot substitute for misdirected resource utilization and poor leadership, as is demonstrated by the fact that fully one-third of all projects fail and another third end up well below expectations.

Projects are conceived for a wide array of purposes, such as product development, construction, enterprise software installation, process re-engineering, or service deployment. While each of these examples exhibit unique characteristics, a capable manager needs to focus on nearly similar constraints of people, money, and performance expectations; and each project requires the project manager to motivate and lead her team members, and to make the appropriate decisions on the costs, tradeoffs, and risks.

The course aims to provide relevant project management skills by exploring project decisions across three modules: (i) structuring and managing the task and leading the project team in an individual project, (ii) aggregate linkages across a portfolio of projects and management of programs, (iii) alliances across firms, project contracting and managing open innovation.

LEARNING OBJECTIVES

MODULE 1

Team design and leadership: Heavy weight project teams, conflict management, designing information flows, modularity, team leadership, matching teams to tasks.

Task design and oversight within a project: stage gate processes, milestones, critical path method, modularity, performance measures, recent views on project success, fast track project management.

MODULE 2

Risk management: valuing real options, project portfolio as options, managing residual risk, building in flexibility, cost of flexibility.

Program and portfolio management: managing to learn, project postmortem, valuation, technology selection and platform planning, product suites, and product architecture

MODULE 3

Distributed project management: contracting in project management, trust and trustworthiness, open innovation, contests and contest architectures.

Management of novel projects: managing chaos, experimentation as a tool, flexibility and learning.

COURSE MATERIALS

A class-reader with a carefully selected set of readings has been designed for this course. In addition to the reader, I may hand out additional reading materials in class.
The assigned readings are given in the course schedule.
Each student is expected to have completed the reading in preparation for each class before the class starts.

ATTENDANCE

You are expected to attend every class; you are responsible for the material covered in class whether or not you attend. I realize that despite your best efforts you might be forced to miss a class due to unforeseen emergencies, and I would like you to inform me if you have (or will miss) a class.
Please avoid using laptops, PDAs, and/or any internet devices while the class is in progress.

TEAM WORK

You shall pick a partner to work with, and submit joint reports on the assigned cases. The goal of this buddy-system is to ensure that everyone in the class is on the same page and to promote critical thinking when writing up the assigned cases. You will select your team member by the end of first week of classes.

EVALUATION AND GRADING

Students will be evaluated based on their performance along the following dimensions:

1. Group case memos (30 pts) –There will be **six** case memos to be completed in groups of two by the due date specified on the attached course schedule. Before the due date, you shall read the case and as a group, write a **single** two page memo answering the assigned question/s. Each is worth 10 team pts. That is, each person can get up to **5 individual pts.**

   It is your responsibility to make sure that you and your team mate submit the case memos before the assigned dates. ***Late submissions shall not be accepted.*** *In case you have any emergency that prevents you from turning in the case memos on time, it is your responsibility to inform me of this promptly.*

2. Exam (1st Milestone): 35 pts - There will be an 120 minute midterm exam (see schedule below). It covers all text readings, cases, homework and class discussions up to that time.

3. Exam (2nd Milestone)35 pts –We will have a second 120 minute final exam. The final exam will **not** be comprehensive. The exam can cover any topics from the text readings, cases, class discussions or homework assignments.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: http://www-senate.ucsd.edu/manual/appendices/app2.htm#AP14

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.

TENTATIVE SCHEDULE

The following is a tentative schedule. In general, even if the specific date of coverage may change slightly, the order of coverage, and the associated material should remain as presented below.

Session	Topics	NOTES	Readings
1	Overview, basics of PM: planning vs. execution, stage-gates		1. What makes projects successful? HBS Press Chapter, HBS Product #2833BC
	working in project teams, classroom exercise		
2	concurrent engineering Fast track project management	#1 Case memo	2. Toyota's principles of set-based concurrent engineering, Sloan Management Review, Winter 1999, Volume 40, No. 2, pp 67-83 3. Jalopy sports car development: managing concurrent engineering projects, INSEAD Case 696-042-1
	Critical path method		
3	Organizing teams and leadership, functional teams, heavy weight teams,	#2 Case memo	4. Eli Lily: Evista Project. HBS Case
	"fit" between team and task, a classroom exercise		5. Manuel E. Sosa, Steven D. Eppinger, Craig M. Rowles, Are Your Engineers Talking to One Another When They Should?, HBR, Nov 01, 2007.
4	More scheduling, managing teams & disruptive projects	#3 Case memo	6. Dragonfly: Developing a proposal for an uninhabited aerial vehicle (UAV), INSEAD Case 600-003-1
	Fast track project management, review		
5	**Midterm exam**		
	Portfolio management		
6	Program Management	#4 Case	7. Mission to Mars (A), HBS Case # 9-603-083

Session	Topics	NOTES	Readings
	learning from failures, project post-mortem Risk management, Real Options: Valuing flexibility	memo	8. Stefan Thomke and Steven Sinofsky. 1999. Learning from Projects: Note on conducting postmortem analysis
7	Portfolio mgmt. exercise		
8	**Debrief**	#5 Case memo	9. Merck & Company: Evaluating a Drug Licensing Opportunity, Richards Ruback, HBS, March 2003., Product # 201023
	Open innovation. Contests and architecture of contests		10. The Ins and Outs of Open Innovation, Raul Chao, HBS Product #UV6666-PDF-ENG
9	Product suites, and managing product architectures	#6 Case memo	11. Stefan Thomke, Steven Jay Sinofsky.,Microsoft Office: Finding the Suite Spot, Product number:699046-PDF-ENG 12. Planning for Product Platforms, David Robertson & Karl Ulrich, HBS Product #SMR037-PDF-ENG
	Contracting, and Behavior alignment through explicit incentives		
10	Managing novel projects: role of experimentation & flexibility Unforeseeability and uncertainty		13. Vol de Nuit - The Dream of the Flying Car at Lemond Automobiles SA, INSEAD Case # 603-006-1
	Review		

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Assignment Questions for Case Memos

You will work with one other classmate to analyze/discuss these cases, and submit a single case memo. In case we have an odd-sized class, we can have one team that has 3 members.

The case memo shall be 1-1/2 spaced, use font-size 12, must not exceed 2 pages. The memo shall be due at the beginning of the class before we discuss the case and should be uploaded on canvas. You may attach additional exhibits or appendices if necessary. Some of the answers may be short and not need more than 3-4 sentences to answer. That is, there is no need to write long answers merely to fill up more pages.

Jalopy Case

1. If you take the delay of the prototype as given (uncontrollable, due to "unplannable" adverse events), what are the fundamental conditions that caused the rework delay in die cutting?

2. Can you identify managerial actions to reduce the die sensitivity to allow earlier prototyping changes?

3. Do you see organizational communication problems that may be have contributed to Jalopy's rework problems, in addition to late prototype changes and die sensitivity?

Eli Lilly Case

1. What is a "heavyweight project team" and how does it differ from the traditional approach used by Eli Lilly?

2. What is your assessment of the performance of two heavyweight teams described in the case? What factors contributed most to these performance results (choose the factors and the performance metrics that you think are most important)? In the pharmaceutical context, how far back in the development process should heavyweight teams be deployed? Why?

3. Is the "heavyweight" project team an approach that you would recommend to Lilly for purposes of commercializing the Evista Project? Why or why not?*When making a specific choice (for question 3), you need to (i) argue for your choice, and (ii) actively think of*

counter-arguments, and how the concerns raised by these counter-arguments can be addressed.

Dragonfly Case

1. How can the project schedule be formally represented, showing interdependence and parallelism among activities (please provide the network diagram)?
2. What is the *critical path* of the project? What completion time and budget does it imply (pretend for now that activity times are deterministic)?
3. How does the project schedule change, when activity time uncertainty is introduced? How does the project budget change? (For this question, assume that the interdependence among activities A4 and A9 does not exist). *(a "reasonable" answer that is in the ballpark is adequate)*

Mission to Mars Case

1. What is your evaluation of the FBC initiative? What were its aims? What did it do well? What, if anything, went wrong with it?
2. Are smaller missions more or less risky than larger missions? What does the case data suggest?
3. What should Naderi do with respect to the 2001 missions?

Merck Case

1. Build a decision tree model that shows the cash flows and probabilities of the stages of the FDA approval process.

2. Should Merck bid to license Davanrik? How much should Merck pay?

3. What is the expected value of the licensing arrangement to LAB? Assume a 5% royalty fee on any cash flows that Merck receives from Davanrik after a successful launch.

4. How would your analysis change if the cost of launching Davanrik for weight loss were $225 million instead of $100 million as given in the case?

Microsoft Office Case

1. What is the state of the marketplace for productivity application suites?
2. How would you describe the current development process and organization?
3. What are the challenges of integrating individual products, which compete in independent categories, into an integrated whole?
4. What should be the contribution of the newly formed Office Product Unit, and how could these goals be best realized?
5. What is the best path for the next release of Microsoft Office: the “12/24” proposal or the “15 month” proposal?